

SLICKTRACE AI

Oil Spill Detection & Evidence Based Vessel Attribution

“Detect the slick. Trace its probable origin. Rank the vessels.”

Problem Statement	SIH26143
Team Name	Indi - Innovators
Project Type	Evidence based investigative decision support prototype
Development Status	End-to-end working MVP with deployed Stream lit dashboard
Live Dashboard	SlickTrace AI
Repository	Githubrepo

1. Executive Summary

SlickTrace AI is an evidence based investigative decision support system for detecting oil spill patterns in Sentinel1 Synthetic Aperture Radar (SAR) imagery and tracing their probable origin toward candidate vessels using historical Automatic Identification System (AIS) evidence. The system is designed around a critical distinction: detecting an oil slick is not the same task as attributing its release to a vessel.

The current working prototype implements an end-to-end pipeline consisting of SAR preprocessing, binary oil spill segmentation using DeepLabV3+ with a MobileNetV2 encoder, spill characterization, a custom backward drift prototype, real historical AIS processing from MarineCadastre, evidence consolidation and candidate vessel ranking, and an investigator oriented Streamlit dashboard.

The system deliberately avoids treating the nearest vessel as responsible or presenting an evidence score as a probability of guilt. Instead, it produces ranked candidate vessels together with supporting spatial and trajectory evidence, while allowing the outcome to remain inconclusive when evidence is insufficient.

CORE IDEA

The system does not replace an investigation. It organizes the evidence chain so that an investigator can see how a candidate reached the ranking and where the evidence is still incomplete.

The deployed dashboard is available at <https://slicktrace-ai.streamlit.app/>.

2. Abstract

SlickTrace AI is an investigative decision support prototype built for SIH26143, aimed at connecting two tasks that are often conflated in oil spill response: detecting a slick, and identifying which vessel may be responsible for it. Given a Sentinel1 SAR scene, the pipeline segments candidate oil spill regions using a DeepLabV3+ model with a MobileNetV2 encoder, characterizes the detected spill geometrically, estimates a probable origin region through a custom backward drift prototype, and cross-references that region against real historical Automatic Identification System (AIS) vessel tracking data sourced from MarineCadastre. The output is a ranked list of candidate vessels supported by spatial and trajectory evidence, not a verdict.

Every stage of the system is built around a deliberate refusal to overclaim. A detected slick is evidence of a pattern, not proof of a cause. A vessel near the estimated origin is a candidate for further investigation, not a suspect. The complete workflow, from SAR input to ranked evidence, is exposed through a Streamlit dashboard deployed for live demonstration, so that an evaluator can walk through the reasoning at each stage rather than receiving a single opaque score.

3. Introduction

Marine oil spills are usually first noticed well after the release event, once the slick has already drifted with wind and surface currents. Investigators are then left working backward: they have an image of where the oil is, not where it came from, and a shipping lane full of vessels that were plausibly nearby at the time. Satellite radar has become a practical tool for the first half of that problem, Sentinel1's Cband SAR imagery does not depend on daylight or clear skies, which is precisely when many spills are least likely to be caught by aerial patrol or optical sensors.

The harder half is turning a detected slick into something an investigator can act on. SlickTrace AI is scoped around the second half. It treats detection as a starting point rather than an endpoint and builds a short evidentiary chain from the SAR scene through geometric characterization, backward drift estimation, and historical AIS cross-referencing, ending in a ranked, not conclusive list of candidate vessels.

4. Problem Statement

An oil spill investigation must answer three separate questions before it is useful to anyone: what was detected, where the release likely occurred, and which vessels were operating near that location around the relevant time. Each question is answered by a different kind of evidence and treating them as one question is where naive approaches tend to fail.

The slick observed in a SAR scene is not necessarily where the spill happened to wind, and current can move surface oil several kilometers from its source within hours. Vessel traffic records for any reasonably busy stretch of water will show dozens or hundreds of ships in the general area; most of them uninvolved. Sorting through this without a structured pipeline leaves an investigator manually cross-referencing satellite images against AIS logs, with no consistent way to decide which vessels warrant a closer look.

SlickTrace AI is built to shorten that manual process combining SAR based detection, geospatial spill characterization, a drift-based origin estimate, and historical AIS filtering into a single ranked output without pretending to resolve legal responsibility along the way.

5. Objectives

Detect oil spill patterns in Sentinel1 SAR imagery using a trained segmentation model.

Characterize each detected spill geometrically centroid and area from the predicted mask.

Estimate a probable origin region for the spill using a backward drift prototype.

Filter historical AIS observations to identify vessels operating near the relevant area.

Combine spatial and trajectory evidence into a transparent, ranked candidate vessel list.

Present the full investigation through a dashboard suitable for live demonstration and decision support.

Keep uncertainty explicit throughout and avoid presenting any output as a determination of vessel responsibility.

6. Existing Challenges

Turning satellite detection into usable investigative evidence runs into a few recurring difficulties, most of which shaped how SlickTrace AI was scoped.

Segmentation on SAR imagery is noisy by nature wind slicks, algal mats, and other “lookalikes” produce dark patches in radar backscatter that resemble oil without being oil, and a model trained on this data has to be evaluated with that ambiguity in mind rather than judged only against clean, favorable examples.

Backward drift estimation depends on accurate environmental forcing wind, currents, and sea state and a prototype without access to validated oceanographic datasets such as CMEMS or ERA5 can only approximate that behavior rather than reproduce it precisely.

AIS coverage, meanwhile, is historical rather than Realtime in most public archives, and satellite acquisition timestamps are not always available with the precision needed to claim that a vessel's position and the spill timing genuinely line up. A system that skips past these gaps and reports a single confident number is, in practice, hiding them rather than solving them which is the gap this project tries to close honestly rather than paper over.

7. Proposed Solution

SlickTrace AI addresses this by keeping each stage of evidence separate and visible rather than collapsing everything into one score. A Sentinel1 scene is segmented for oil spill patterns, the resulting mask is converted into geospatial spill characteristics, a backward drift prototype estimates where the spill probably originated, and that origin region is cross-referenced against real historical AIS traffic to shortlist and rank candidate vessels. Every stage is shown to the investigator individually through the dashboard, so the reasoning stays auditable rather than hidden behind a final probability figure.

The prototype currently favors a working, honestly scoped pipeline real detection, real historical AIS, a functioning backward drift approximation over an unfinished attempt at a fully operational oceanographic simulation.

Principle	Meaning in SlickTrace AI
Detection ≠ Attribution	A detected slick is evidence of a possible spill pattern, not evidence that a particular vessel caused it.
Spill location ≠ Release location	The observed slick may have moved after release.
Nearest vessel ≠ Responsible vessel	Proximity is only one piece of evidence.
Backward drift ≠ Exact source proof	The prototype estimates a probable origin region, not a legally exact release point.
Evidence score ≠ Probability of guilt	The ranking is an investigative prioritization mechanism.
Candidate vessel ≠ Responsible vessel	A candidate requires further investigation and corroboration.
No high confidence match is valid	The system should not force an attribution when evidence is insufficient.

8. System Architecture

The pipeline runs as nine sequential stages: SAR input, preprocessing, AI based oil spill detection, spill characterization, backward drift estimation, historical AIS processing, spatial and trajectory evidence construction, candidate vessel ranking, and finally the investigator dashboard that ties the stages together.

PIPELINE

Sentinel1 SAR → preprocessing → segmentation → spill characterization → backward drift → historical AIS → spatial / trajectory evidence → candidate ranking → investigator dashboard.

This differs in one respect from the architecture originally planned for the project. The initial design called for operational drift simulation using OpenDrift/OpenOil, driven by environmental forcing from CMEMS and ERA5. Neither integration is complete, and this report does not present them as working components in the backward drift stage currently in place is a custom prototype, described in Section 9.5, and CMEMS/ERA5 forcing remains for future work.

9. Methodology

9.1 SAR Data

The project uses the Sentinel1 SAR Oil Spill Dataset, structured into three parts of oil, lookalike/no oil, and test. Verified source imagery is 2048×2048 pixels across two SAR bands (VV and VH, float32), with binary [0, 1] masks.

Dataset component	Verified use
Part I - Oil	Core oil spill training data.
Part II - Lookalike / Nooil	Negative and lookalike examples used in the training subset.
Part III - Test	Model inference / evaluation and geospatial demonstration.

After filtering and verification, the training subset used for the current model contained 1,799 samples including 479 lookalike and 480 no oil examples alongside the oil positive samples with a separate validation set of 385 samples. Image and mask counts were checked against each other during dataset verification, and this check passed without a mismatch.

9.2 Preprocessing

Each SAR scene is read with Rasterio and prepared as a two band NumPy array (VV and VH) before being passed to the segmentation model. This stage is kept deliberately lightweight; no attempt is made to fuse additional bands or auxiliary layers beyond what the trained model expects.

9.3 DeepLabV3+ Segmentation

Oil spill detection is framed as binary semantic segmentation: DeepLabV3+ with a MobileNetV2 encoder predicts, for every pixel in the input scene, whether it belongs to an oil spill region. A MobileNetV2 backbone extracts features at progressively coarser scales, an Atrous Spatial Pyramid Pooling (ASPP) module aggregates multiscale context at the bottleneck, and a lightweight decoder up samples the result while fusing in a low-level skip feature from early in the encoder.

Training parameter	Verified configuration
Input size	2048 × 2048
Input channels	2 (VV + VH)
Batch size	1
Learning rate	0.0001
Weight decay	0.0001

Loss	BCEWithLogitsLoss + DiceLoss
Metrics	Dice, IoU, Precision, Recall
AMP	Enabled
Workers	0

9.4 Spill Characterization

Once the model produces a predicted mask, the pipeline converts it into a spill event to record the predicted spill pixels are used to derive a centroid and an area square kilometers. For the case used throughout this report, the predicted spill is centered at 28.942782, -88.834958 with an area of 57.677181 km², in the Gulf of Mexico.

INTERPRETATION

This case is used purely as a working demonstration scene. It is not evidence of, and should not be read as, a real oil spill incident in Indian waters, nor as an accusation against any named vessel.

9.5 Backward Drift

Starting from the characterized spill location, the backward drift prototype steps backward through a fixed time horizon to estimate where the spill probably originated. For the demonstration case, the prototype ran 25 steps at 0.5hour intervals on a 12hour backward horizon and arrived at an estimated probable origin of 29.025164, -88.929210, with a 5 km uncertainty radius.

This is a probable origin region, not an exact release point, and the distinction matters how the output should be read. The drift model implemented here is a custom prototype; it is not yet driven by validated environmental forcing. Operational hindcasting with OpenDrift/OpenOil, using wind and current data from CMEMS and ERA5, remains a planned extension and is not represented as working in this pipeline.

9.6 Historical AIS Processing

Vessel traffic evidence comes from real historical AIS data obtained through MarineCadastre, not from synthetic or simulated tracks. For the demonstration case, the processed AIS source is dated 20250108 and contains 342 unique vessels. A broad 50 km discovery radius is applied first, followed by a tighter 10 km primary proximity filter, which narrows the pool to 96 primary proximity candidates for further evidence analysis.

Temporal compatibility between the AIS records and the SAR acquisition is not claimed for this scene, because a verified satellite acquisition timestamp is not available for it. Where that timestamp is missing, the dashboard reports temporal matching as unavailable instead of estimating a time alignment it cannot support.

9.7 Evidence Based Vessel Ranking

The remaining 96 candidates are scored using spatial proximity and trajectory related features and consolidated into a ranked list intended to help an investigator prioritize which vessel records deserve closer examination first.

The intended reading of this output is deliberately narrow. A candidate vessel is not a responsible vessel: a high evidence score means the available prototype evidence is more relevant under the current scoring logic, not that guilt is more likely. The ranking's job ends at investigator reviews. Everything after that point is human judgement, not model output.

10. Implementation

Table 4 summarizes the technology choices behind each pipeline stage.

Component	Technology	Current role
SAR data processing	Rasterio, NumPy	Read and prepare two band SAR imagery.
Segmentation model	PyTorch, segmentation_models_pytorch	Binary oilspill segmentation.
Model architecture	DeepLabV3+ + MobileNetV2	Pixel level oilspill prediction.
Geospatial processing	GeoPandas, Shapely, Rasterio	Spill geometry, centroid, and area characterization.
Drift prototype	Custom Python backwarddrift model	Estimate probable origin region.
AIS processing	Python / pandas / geospatial analysis	Historical vessel filtering and evidence analysis.
Dashboard	Streamlit, Folium, streamlitfolium	Investigator facing visualization and evidence presentation.
Deployment	Streamlit Community Cloud	Public demonstration of the lightweight prototype.

Public deployment does not ship to the full training environment. It uses a separate, lightweight dependency set Streamlit, pandas, Folium, streamlitfolium, and Rasterio packaged with a representative demo data bundle containing one SAR scene, its groundtruth and predicted TIFFs, evaluation metrics, spill event data, backwarddrift output, and AIS evidence results. Heavier local resources, including the 78.07 GB training subset, the raw AIS archive, and the full set of model checkpoints, are intentionally excluded from the public package, so the dashboard stays deployable on Streamlit Community Cloud.

11. Experimental Results

The Part III aggregate evaluation result at a 0.55 prediction threshold is the primary set of figures used to describe measured model performance.

Metric	Measured result
Dice	0.6601
IoU	0.6261
Precision	0.7032
Recall	0.6538

HOW TO REPORT THE RESULT

These four values are the aggregate Part III result. A separate 12scene gallery was curated from favorable oil positive examples; its Dice range of 0.94300.9749 is useful for demonstration, but it is not representative of overall performance.

The implementation status at the time of this report is as follows:

Pipeline stage	Status
Sentinel1 SAR input	Implemented
SAR preprocessing	Implemented
DeepLabV3+ oil spill segmentation	Implemented and tested
Spill characterization	Implemented / prototyped
Backwarddrift prototype	Implemented
Operational OpenDrift/OpenOil hindcasting	Not completed
CMEMS / ERA5 environmental forcing	Not completed
Real historical AIS processing	Implemented
Spatial / trajectory evidence	Implemented
Candidate vessel ranking	Implemented
Streamlit investigator dashboard	Implemented and deployed

12. Dashboard

The deployed dashboard, built with Streamlit and Folium, walks an investigator through the pipeline as a sequence of evidence views rather than a single Blackbox prediction.

Dashboard capability	Purpose
SAR Input	Display the selected SAR scene and input imagery.
Detection / Model Evidence	Show model prediction evidence and measured metrics.
Spill Characterization	Present spill geometry and geospatial measurements.
Backward Drift	Show the estimated backward trajectory and probable origin region.
AIS / Vessel Evidence	Present candidate for vessel evidence and ranking.
Investigation overview	Connect the evidence stages into an investigator-oriented workflow.

LIVE DEMONSTRATION

The deployed dashboard is available at <https://slicktrace-ai.streamlit.app/>.

13. Case Demonstration

To make the pipeline concrete, this section walks through the single demonstration case used throughout the report, tying together the numbers introduced separately in the methodology.

The input scene is a Sentinel1 SAR image over the Gulf of Mexico. Segmentation identifies a spill region centered at 28.942782, -88.834958, covering approximately 57.68 km². Running the backward drift prototype over a 12hour horizon (25 steps, 0.5hour intervals) places the estimated probable origin at 29.025164, -88.929210, with a 5 km uncertainty radius. Filtering the 20250108 historical AIS archive 342 unique vessels first with a 50 km discovery radius and then a 10 km primary proximity filter narrows the field to 96 primary candidates, which are then scored and ranked as spatial and trajectory evidence.

Parameter	Value
Region	Gulf of Mexico
Spill centroid	28.942782, -88.834958
Spill area	57.677181 km ²
Estimated probable origin	29.025164, -88.929210
Uncertainty radius	5 km
Drift horizon	12 h (25 steps × 0.5 h)
AIS source date	20250108
Vessels processed	342
Primary proximity candidates	96

CASE BOUNDARY

This is a Gulf of Mexico prototype scene used to demonstrate a working pipeline not a real-world Indian spill investigation, and not an accusation against any specific vessel.

14. Limitations

The demonstrated drift model is a custom prototype and is not yet an operational oceanographic hindcast.

CMEMS and ERA5 forcing have not been integrated into the working simulation.

OpenDrift/OpenOil is a planned extension rather than a completed production component.

Temporal AIS compatibility cannot be claimed when a verified satellite acquisition timestamp is unavailable.

SAR TIFF georeferencing quality varies; some source files generated a Rasterio NotGeoreferencedWarning during inspection.

The selected Gulf of Mexico case is a prototype demonstration and is not evidence of a real Indian incident.

The vessel ranking does not establish legal responsibility or guilt.

The selected 12-scene gallery is curated for demonstration and must not be treated as overall model performance.

15. Future Scope

Integrate operational OpenDrift/OpenOil hindcasting.

Use validated environmental forcing from CMEMS and ERA5.

- Improve uncertainty estimation for probable origin regions.
- Add verified satellite acquisition timestamps and rigorous temporal AIS matching.
- Extend vessel evidence with richer trajectory and behavioral features.
- Evaluate the system on additional, geographically diverse, and independently verified spill cases.
- Introduce stronger provenance and audit trails for investigation records.
- Improve deployment scalability while preserving evidence of transparency.

16. Conclusion

SlickTrace AI demonstrates an end-to-end prototype for oil spill investigation that connects satellite-based detection, geospatial spill characterization, probable origin estimation, historical AIS analysis, and evidence-based candidate vessel ranking in a single investigator facing workflow.

The central contribution is the separation of detection from attribution. Rather than treating a segmentation result or vessel proximity as proof of responsibility, SlickTrace AI organizes multiple evidence sources and presents candidate vessels for further investigation. This matters most in exactly the cases that are hardest to reason by eye where the observed slick has drifted away from its release location, or where several vessels operated in the surrounding region at once.

The prototype is intentionally transparent about its own boundaries. The backwarddrift stage is a prototype, operational environmental forcing remains future work, and the vessel ranking is not a guilt determination. Within those limits, the deployed dashboard provides a working demonstration of the complete investigative workflow, built and evaluated honestly against what it can and cannot yet claim.

SLICKTRACE AI

Detect the slick. Trace its probable origin. Rank the vessels.

17. References

- MarineCadastre Historical AIS data access: <https://marinecadastre.gov/accessais/>
- Sentinel1 SAR Oil Spill Dataset project dataset source referenced in the project artifacts.
- SlickTrace AI GitHub repository: <https://github.com/alina-sayyad/SlickTrace-AI>
- SlickTrace AI live Dashboard: <https://slicktrace-ai.streamlit.app/>

18. Appendix

Appendix A Judge Demonstration Flow

- Open the live SlickTrace AI dashboard.
- Show the SAR input scene and explain that the model operates on VV/VH SAR channels.
- Show the segmentation/model evidence and be clear about the difference between the aggregate Part III result and the curated demonstration gallery.
- Open spill characterization and point out the spill centroid and area.

- Open backward drift and explain that the output is a probable origin region, not an exact release point.
- Open AIS evidence and explain that real historical MarineCadastre AIS is used.
- Show the candidate vessel ranking and explain why it is evidence of prioritization rather than a responsibility determination.
- Close by stating the current limitations and the future operational hindcast scope.

Appendix B Terminology to Use During Presentation

Use	Avoid
Probable origin region	Exact source location
Candidate vessel	Responsible vessel
Evidence ranking	Probability of guilt
Historical AIS evidence	Realtime AIS, unless available
Backwarddrift prototype	Operational oceanographic hindcast
Selected demonstration scenes	Representative overall performance
Investigative decision support	Automated legal attribution

Appendix C Quick Facts

Item	Current project value
Model	DeepLabV3+ + MobileNetV2
Input	2048 × 2048, VV + VH
Training subset	1,799 samples
Validation set	385 samples
Part III Dice	0.6601
Part III IoU	0.6261
Part III Precision	0.7032
Part III Recall	0.6538
Demonstration spill area	57.677181 km ²
Backward horizon	12 hours
AIS source	MarineCadastre, 20250108
AIS vessels processed	342
Primary proximity candidates	96